

GIT AND GITHUB

Git is an open source and distributed version control system and it follows 3 tier architecture

- Working directory
- Local Repository
- Remote Repository

GIT COMMANDS

- ✓ **To check the git is downloaded**
 - `git --version`
- ✓ **To download the git repo to your folder**
 - `git clone git-repolink`
- ✓ **To create a connection between local repository and remote repository**
 - `git init`
 - `git remote add origin git-repolink`
- ✓ **To Config in windows:**
 - `git config --global user.name=""`
 - `git config --global email=""`

Staging (add):

- ✓ **To add a single file**
 - `git add filename`
- ✓ **To add more than 1 file**
 - `git add filename1 filename2`
- ✓ **To add all files**
 - `git add .`
- ✓ **To unstage a single file**
 - `git rm --cached filename`
- ✓ **To unstage all files**
 - `git reset`

Local Repository (commit):

- ✓ **Send files to Local Repo:**
 - `git commit -m "Your commit message"`
- ✓ **Back file to staging:**
 - `git restore --staged filename`

Remote Repository (push/pull):

- ✓ **Local Repo to Remote Repo:**
 - `git push origin branch_name`

- ✓ **From Remote to Local Repo:**
 - git pull origin branch_name
- ✓ **Set Upstream**
 - git branch --set-upstream-to=origin/branch

Branching:

- ✓ **Create new branch:**
 - git branch new_branch_name
- ✓ **Select new branch:**
 - git checkout new_branch_name
- ✓ **Create and select new branch:**
 - git checkout -b new_branch_name
- ✓ **Rename branch**
 - git branch -m new_branch_name (**For current branch**)
 - git branch -m old_branch_name new_branch_name (**For another branch**)
- ✓ **Delete branch:**
 - git branch --delete branch_name

Note: You cannot delete the current_branch
- ✓ **Move new branch to github:**
 - git push origin new_branch_name
- ✓ **To See how many branches in Remote Repository:**
 - git branch -a
- ✓ **Merge new_branch with main/master:**
 - Go to main/master -> git checkout main/master
 - **Merge:** git merge branch_name
- ✓ **To Check Current branch:**
 - git branch

Common Commands:

- ✓ **To Check Commit Log:**
 - git log
- ✓ **To Check status:**
 - git status. It shows about the untracked fields, ready to be committed, modified files
- ✓ **To Check the modification**
 - git diff
- ✓ **To merge only the particular commit:**
 - git checkout main/master
 - git cherry-pick commit_id (use git log for commit_id)

Scenario:

When you pull from remote repo, the local repo file not matching means, it gives conflict and we need to decide whether to commit or stash(move local files to temporary memory and after receive the remote repo file and then we want to merge local and remote file that we are pulled, then also it shows conflict, then go to conflicted file and see the upstream and stash changes which change do you want and save the file)

- ✓ **To move the conflict local file to temporary memory (Stash)**
 - git stash
- ✓ **To get the conflict file from stash memory**
 - git stash pop